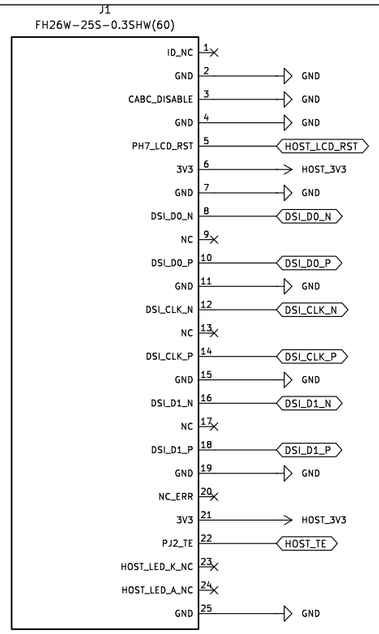
01 / STM32F469I-DISCO > DISEN DS070HSD39T-132

MIPI DSI / LCD interface | Host CN10: UM1932 Rev 5 tables 6–7 | Panel: DISEN A0 preliminary

01 / STM32 HOST – CN10



25-pin FPC / 0.30 mm pitch

HOST CONNECTION NOTES

Remove original LCD and touch flexes.
Use purpose-made 1:1 FPC cables.

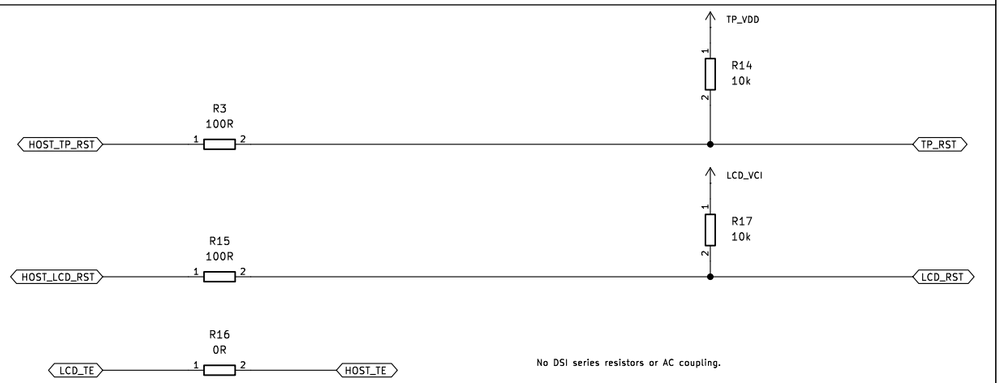
J1.23 / J1.24: host LED outputs unused.
J1.3 = GND disables default CABC control.
UM1932: R117 fitted / R119 absent.
Check actual host revision and population.

REV J / LAB PROTOTYPE – Electrical review; not a manufacturing release.
Panel current, flex mating and OVP transient qualification remain open; see review report.

02 / MIPI DSI – DIRECT PAIRS

NET / PAIR	HOST J1	LCD J2
DSI_D0_P	10	16
DSI_D0_N	8	17
DSI_D1_P	18	20
DSI_D1_N	16	21
DSI_CLK_P	14	24
DSI_CLK_N	12	25

03 / RESET & FRAME SYNCHRONIZATION



No DSI series resistors or AC coupling.

05 / POWER, TOUCH & BACKLIGHT SHEETS

Logic power & touch

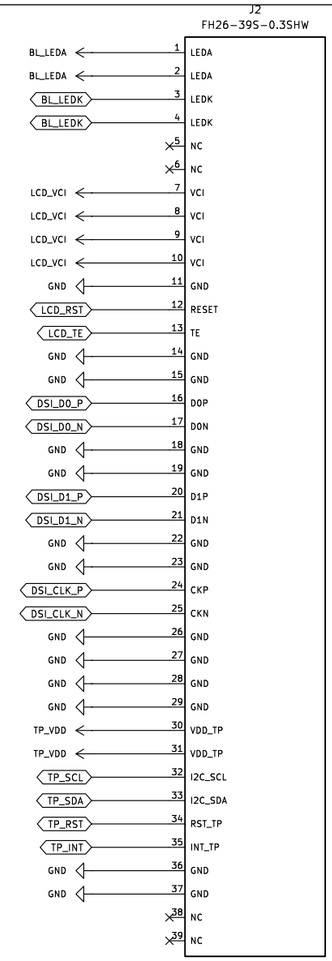
File: Logic_Touch.kicad_sch

Backlight / external power

File: Backlight.kicad_sch

Logic: host 3.3 V only. Boost: separate regulated 5 V.
All blocks share GND. Identical net labels are connected.

04 / DISEN LCD – 39-PIN FPC



0.30 mm pitch / verify contact side & pin 1

DSI: 2 data lanes + clock, direct DC coupling.
100 ohm nominal differential; continuous GND.
OTA7290B / ILI2130 are panel devices.
J2 corrected to FH26; verify physical flex mating.

Lab prototype | Native electrical review | See Engineering review

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Sheet: /
File: STM32F469_DISEN_RevJ.kicad_sch

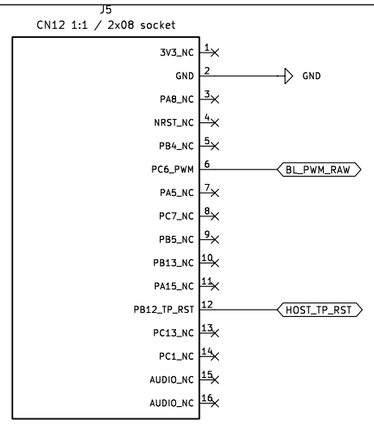
Title: STM32F469 / DISEN – MIPI & LCD interface

Size: A3	Date: 2026-10-16	**Rev: J / Electrical review**
KiCad E.D.A. 9.0.2		Id: 1/3

02 / LOGIC POWER, TOUCH & HOST CONTROL

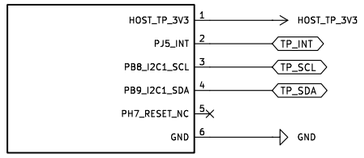
3.3 V logic domains | LCD and touch reset kept independent | Test points grouped by function

01 / STM32 CONTROL — CN12



2x8 socket / 2.54 mm / 1:1 mapping
Follow physical pin 1 for socket orientation.

02 / TOUCH HOST — CN2



6-pin FPC / 0.50 mm pitch
J3.1 supplies TP_VDD through R5.
J3.5 unused: LCD and touch resets independent.

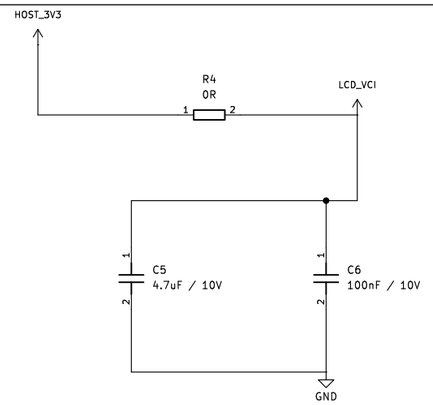
INTERFACE CONVENTIONS

Host LCD 3.3 V and host touch 3.3 V
remain distinct input nets.

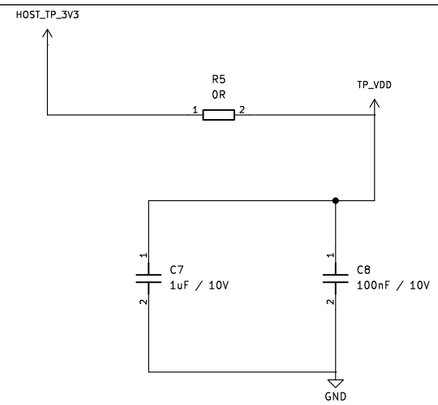
BL_PWM_RAW: PC6 from CN12.
HOST_TP_RST: PB12 from CN12.
TP_INT: PJ5 from CN2.

Power symbols retain original rail names.

03 / LOGIC SUPPLIES & DECOUPLING



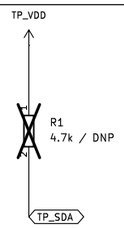
LCD_VCI / host LCD 3.3 V

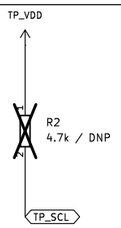


TP_VDD / host touch 3.3 V

Keep 5 V away from both logic rails. Current budget and voltage tolerance require validation.

04 / TOUCH I2C — OPTIONAL PULL-UPS



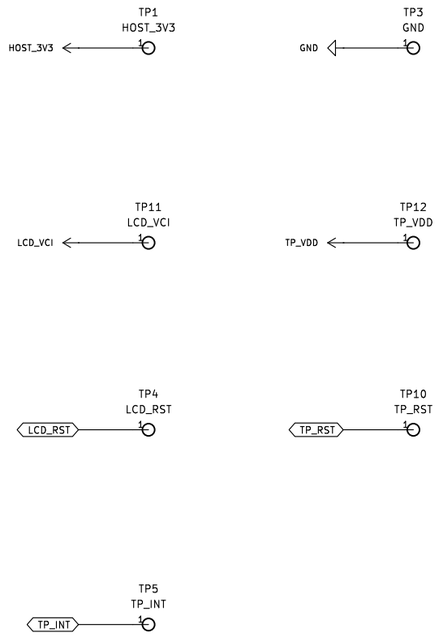


PB8/PB9 = I2C1; PJ5 = INT (official ST BSP).
R1/R2 remain DNP: confirm host pull-ups and measure rise time.
ILI2130 address, interrupt/reset timing need vendor data.

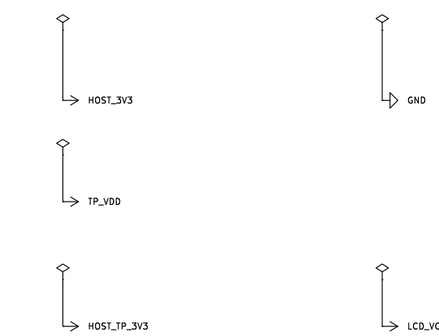
VALIDATION NOTES — REVISION J

LCD VCI: 3.0–3.6 V; touch: 3.3 V nominal.
Drawing specifies touch 2.8–3.3 V: confirm tolerance.
LCD and touch currents are TBD: measure host load.
RESX/ILI2130 timing not specified sufficiently in panel PDF.

05 / LOGIC TEST POINTS



Supply declarations



Lab prototype | Native electrical review | See Engineering review

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Sheet: /Logic power & touch/
File: Logic_Touch.kicad_sch

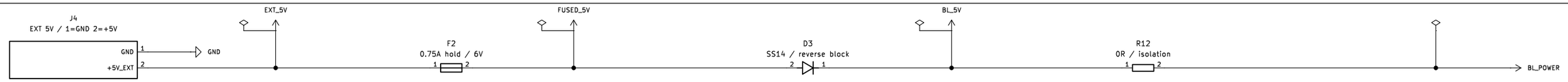
Title: STM32F469 / DISEN — Logic power & touch

Size: A3	Date: 2026-10-16	**Rev: J / Electrical review**
KiCad E.D.A. 9.0.2		Id: 2/3

03 / EXTERNAL 5 V & TPS61165 BACKLIGHT

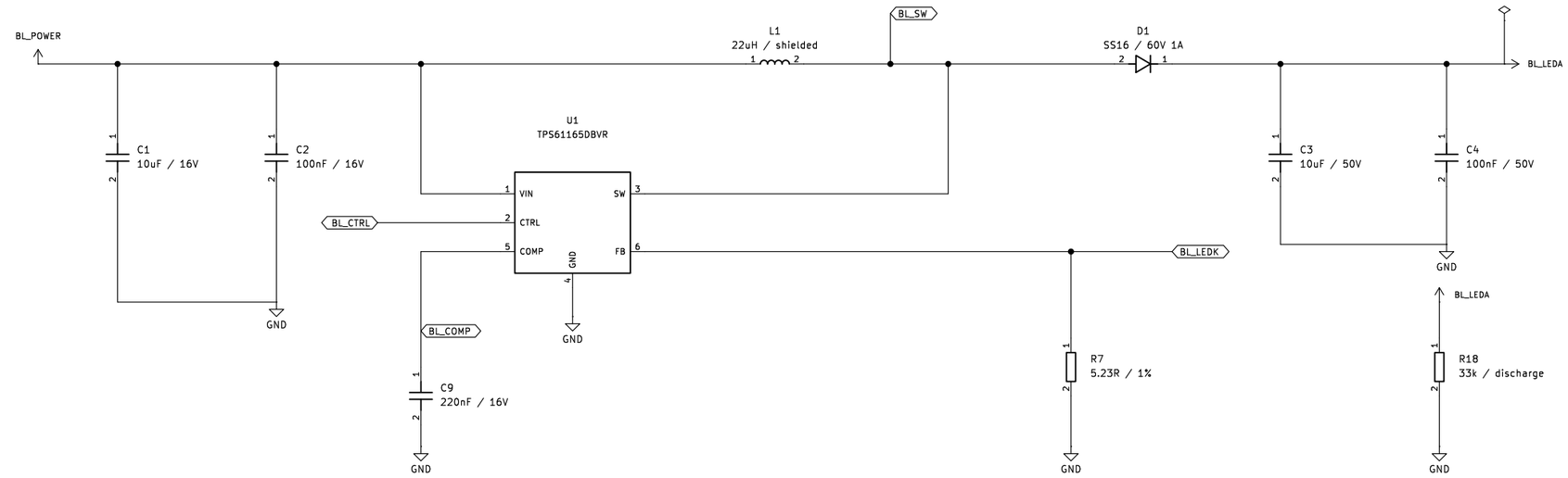
LED load: 21.6–27.2 V at 40 mA | Separate regulated boost supply | Common GND with F469

01 / INPUT POWER – FUSE, REVERSE BLOCK & ISOLATION



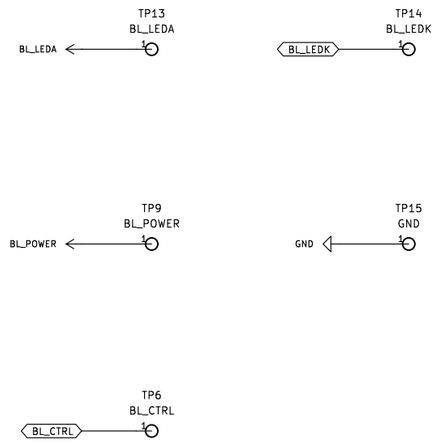
External regulated 5 V is mandatory. J4.1=GND / J4.2=+5V. No connection to host +5 V or host LEDA/LEDK.

02 / BOOST POWER STAGE & LED CURRENT RETURN

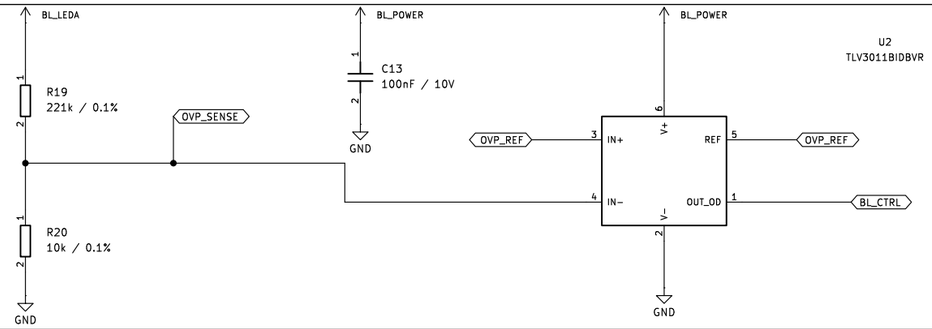


I_LED = 0.200 / 5.23 = 38.24 mA nominal; worst-case full-scale = 39.40 mA (TI VFB + R7 tolerance).

03 / BACKLIGHT TEST POINTS

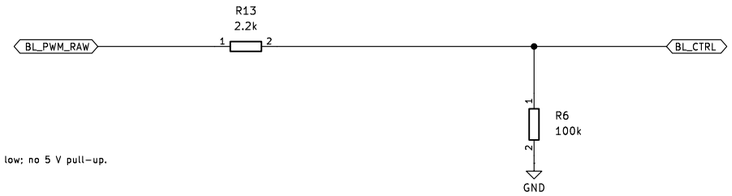


04 / OVP SENSE & COMPARATOR SHUTDOWN



REV J: U2 + C13 now follow BL_POWER; OVP no longer depends on HOST_3V3. Nominal threshold = 28.690 V.
TPS61165 CTRL uses a 180 us filter; full shutdown needs LOW >2.5 ms. This is NOT a hard voltage clamp.
Panel PDF gives LED Vf <=27.2 V at 40 mA; no absolute LED voltage rating or guaranteed 30 V ceiling is provided.
Verify OVP startup/open-load peaks, no-load PWM and power sequencing on the bench. PWM range: 5–100 kHz.

05 / PWM CONTROL & DEFAULT-OFF



Open-drain U2 pulls CTRL low; no 5 V pull-up.

Lab prototype | Native electrical review | See Engineering review

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Sheet: /Backlight / external power/
File: Backlight.kicad_sch

Title: STM32F469 / DISEN – External 5 V & backlight

Size: A3	Date: 2026-10-16	**Rev: J / Electrical review**
KiCad E.D.A. 9.0.2		Id: 3/3